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Willingness-to-pay for improved cyclone early warning services across coastal Bangladesh: Application of choice experiment

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ABSTRACT

Effective early warning services are a prerequisite for significantly minimizing the personal injury, losses of lives and properties from devastating natural hazards like cyclones and storm surges across coastal Bangladesh. This study fills a gap in the literature regarding the value associated with cyclone early warning services. We measure willingness-to-pay (WTP), consumer surplus (CS) and revenue stream in response to the policy change of cyclone early warning services (EWS) on a sample ($n = 219$) observations. Following stratified sampling method, the survey and choice experiment (CE) were conducted in a few coastal villages of four coastal districts of Bangladesh for eliciting stated preference (SP) data. Every participant in the survey faced three options in each card-two hypothetical alternatives and one status quo option. Our proposed attributes for EWS such as accuracy of mean track error, advance update information, and cyclone warning through mobile phone-based short message warning and annual payment for the warning services are considered to construct choice cards. Estimated results ensure that age, family size, years of schooling are the dominating contributors to choose the attributes of EWS. Results of MWTP, WTP, CS, and revenue stream for improved cyclone EWS make a guarantee that coastal households and investors get more benefits and return from improved EWS programs.

1. Introduction

Cyclone warning services are a part of an effective measure of adaptation that received greater attenuation as a powerful tool for disaster risk reduction (DRR) in terms of life saving and asset protection from unwanted catastrophic events like cyclones and storm surges. But the degree of preference to cyclone warning services is not homogeneous in every society. People in high-income countries frequently and digitally get cyclone warning or weather forecast services. But such services do not get popularity in the society of low-income countries [1]. The lower popularity of early warning services (EWS) among targeted populations and regions is a principal issue for the governments in these countries. The evidence from observational studies identify that insufficient mobile phone-based message warning option, shorter coverage area, unaffordability for cyclone warning services, inaccuracy of tropical cyclone forecast, and inadequate update per day, undesirable package of warning services might be bottlenecks leading to low of recipients of EWS [2–8]. These studies predict that among all the problems, mobile phone-based message option, cyclone forecast, and number of updates should be prioritized for improvements to promote EWS. To design significant, effective, and relevant measures of EWS, it is essential to encourage the number of recipients, and policy makers to consider those factors responsible for influencing the targeted populations' demand for better EWS. Beyond social construct, bio-physical thresholds can restrict the benefits of prioritized services of cyclone warning within the boundaries of a particular region. Effects of these services on values of tropical cyclone warning have been reported by Velden et al. [9]; Majumdar et al. [10]; Bhaskaran, Rao, and Murti [11]; Henderson et al. [2]; among others, some have shown the functional relationship between economic value and cyclone warning services as in the case of Uddin, Haque, and Walker [12]; Ferreira et al. [13]; Georgeson, Maslin, and Poessinouw [14], and Lopez et al. [15]. (see Fig. 1)

Economic value placed on cyclone warning services is not homogeneous. This can be induced by direct, indirect, and prospective options [16]. Likewise, passive and non-use motivations also work as significant catalysts to promote cyclone warning services in a society [17]. Socio-economic-demographic (SED) characteristics (i.e., income, age, and schooling years), level of awareness, social desirability, preference, and interest are the influential factors of motivation underlying value for cyclone warning services [18,19]. Economic value may be interpreted

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by public awareness and knowledge of cyclone warning services [20]. However, the following questions remain: how should EWS of cyclones be designed to meet the preferences of coastal people? What is the potential value of willingness-to-pay (WTP) for improved tropical cyclone warning service? Values of WTP measure the ability to pay for improved cyclone EWS. These values could be used to derive values for specific improvement options. The objective of our study is to broaden the scope and rigor of economic valuation for improved cyclone EWS through MWTP, WTP, CS, and Present value (PV).

Quantitative narratives of EWS across coastal Bangladesh are not answered to these questions in terms of market mechanism and are often less persuasive in decision making than quantitative valuation. While translating the value of EWS into further explicit pecuniary terms is highly advocated, it claimed essential knowledge of EWS and preferences of coastal people and their socio-economic-cultural practice for decision making [21]. It also requires sound methodological ground that is effectively valuing of EWS. Our study develops significant strategies for EWS based on perceptions and preferences of coastal people and measures the MWTP, WTP and CS. To the best of our knowledge, the values provided in our study may be the first estimate economic benefits associated with further developments to cyclone warning services not only in coastal Bangladesh but also in similar coastal regions.

Economic valuation for improved cyclone warning service yields direct benefits assessed using stated preference (SP) data through choice experiments (CEs) with a sample of the population of coastal Bangladesh. The revealed preference data (RP) represents data collected on choices that are made in an actual market. Our proposed attributes for improved EWS are a hypothetical concept in coastal Bangladesh. As a consequence, we considered the SP data because of its suitability to choice derived from a hypothetical situation not from an actual market.

Other associated attributes and variables with EWS might also be explained for better empirical assessment. We applied CE approach for measuring the economic values of improved cyclone warning services. It is an effective approach for valuing multi-related fatalities and property damage, cyclone warning services and peace of mind resulting from the rise in safety for both present and future generations [22]. It is an essential tool for WTP. Findings of the article enable to make more specific policies for significant cyclone forecasting and warning dissemination essential to build a calamity resilient coastal society in Bangladesh. Next we offer a description of vulnerability in coastal Bangladesh and government initiatives, highlight our methods to elicit values for the improved cyclone EWS, discuss our results, and offer a summary of our main findings.

Fig. 1. Geographical locations of the survey areas.
Source: [63].

2. Vulnerability to cyclone in coastal Bangladesh and government initiatives

Bangladesh is ranked 7th out of top 10 countries for higher exposure to extreme weather events [23]. Among all extreme events, cyclones are the most devastating [24]. Over the last 100 years, 508 cyclones badly affected the coastal region [25,26] where more than 15 cyclones (covering different category like 2, 3, and 4 guided by the Saffir-Simpson scale at wind speed ranging from 140 to 225 km/h) were more devastating [27]. The Bhola cyclone in 1970 is the most devastating cyclone in world history where three to five hundred thousand people died [28]. In 1991, another devastating cyclone Gorky hit Bangladesh and took about 140,000 peoples' lives and property [29]. During the period of 1980–2000, more than 119 million coastal people are found to be exposed on average every year to tropical cyclone-induced natural hazards [30]. In this period, Bangladeshi coast faced severe cyclones and floods [31]. Bangladeshi coast faces at least one cyclone in every three years at heterogeneous categories [32,33]. Hoque, Pradhan, Ahmed, and Roy [34] report that coastal inhabitants of Bangladesh are considered to be most vulnerable to cyclonic surge.

Position, timing of landfall, intensity, force, and moving path of cyclone and storm surge's height with stage specification during the landfall are essential to design the accurate EWS [35,36]. EWS in Bangladesh disseminated by the Bangladesh Meteorological Department (BMD) is not effective, reliable and significant because of its narrow scope to utilize modern scientific devices with border coverage issues [37]. Currently, BMD uses the traditional method for weather and cyclone forecasting from its 35 weather stations but it needs 500 weather stations for proper and broader weather prediction [38]. Coastal people get BMD disseminated EWS through radio, television, and signal flag demonstration. It is not possible to save lives and property of villagers of remote villages and isolated islands and fishermen of river and off-shore inclusively and equitable from the devastating cyclones and storm surges by these traditional EWS.

The Bangladesh government has recently initiated the project valued at US$ 113 million for strengthening BMD's early warning capacity by supporting early warning generation equipment, weather prediction modeling software, required infrastructure, requirement, and training [39]. Nevertheless, this investment may not be sufficient to minimize the present risks of cyclones and develop the EWS in Bangladesh. Projected adaptation cost for improved EWS requires more projects valued at USD 5516 million and USD 112 million in annual recurrent cost by 2050 [40]. Marketization, perception and preference-based EWS are highly required for proper modernizing and popularizing these services. Payment mechanism of the marketization process ensures the quality of services such as per day updates, mobile phone-based message warning, coverage people, time and location of landfall accuracy with possible impacts, wind speed accuracy and more investment.

Improved cyclone EWS may be one of the leading contributors to achieve the goal of 15 (life on land) of SDGs by 2030. Bangladesh would need to extend coverage to more coastal people, offer more EWS, and pay a greater share of the cost for implementation of improved cyclone EWS. To this end, this study focuses on designing the EWS package and we look forward to potential ways in which the number of recipients can be increased given the quality of desirable cyclone EWS.

3. Methods

3.1. Choice experiment

Choice is an integrated part of human beings [41,42]. It is derived from habit and customs with usual practice and fresh decisions are derived from past experience and present inquiry [43]. Since the early 1970s, human choice has gained more popularity in the field of behavioral economics and successfully applied in the quantitative statistical techniques and other empirical research works [44]. Giving importance on understanding basic features of choice and forecasting the future choice responses, a set of empirical works have evolved. For instance, Gschwandtner, Jang, and McManus [45]; Kim, Kim, & Yoo [46]; and Liski, Koetse, & Metzger [47] synthesize the contributions to the field of choice experiments (CEs).

The CE is well-rooted in welfare economics followed by the concept of neo-classical economic value and maximization of individual utility. Economists, researchers, and policymakers prefer to apply this technique for designing effective and reliable policies because of its versatile and complete methodological ground for benefit estimation in the case of environmental improvements and other public goods [48]. This technique became very popular for valuing non-market environmental goods through WTP followed by the stated preference (SP) data [49]. The SP data and CE are a complement to each other. The SP data appear to date back to the early 1960s when Davis conducted a study in 1963 and highlighted the value of recreation of the Marine wood of the United States [49]. The goal of estimating parameters from dichotomous choice derived from CE response is to measure WTP [50]. Respondents are asked about WTP for a proposed policy through a set of choice cards. A set of choice cards is the major elicitation format of WTP. To reach a certain choice, every respondent must have faced a set of alternative options in a particular set of cards. It comprises at least three alternatives where one must evaluate at least two alternatives to be able to make a specific choice [41]. In exploring the significant choices, an individual faces the range of alternatives and thinks about what influences the decision to choose one alternative over another. These influences are known as attributes induced by prejudice, experience, taste, pay ability, and others and linked to SED characteristics [51]. Such SED characteristics may influence decision making for payments and improved cyclone EWS.

In an experiment of choice, respondents are given hypothetically better or improved options and request them to choose preferred options from different alternatives and levels. Under this experimental technique, respondents are generally requested to maintain a sequence of such choices. When the choice task is finished, they implicitly make trade-offs between the levels of preferred attributes in the different alternatives presented in a choice set [52]. Behavioral framework of the Random Utility Theory (RUT) is the main building block of the choice experiment [53]. It helps to describe discrete choices in a utility maximizing theory. Under this theory, the utility function for a representative consumer can be decomposed into observable and stochastic segments:

$$U_{in} = V_{in} + \in_{jn} \tag{1}$$

where V_{in} shows the systematic and observed factor of the latent class utility and $\in_{in}$ indicates the stochastic unobserved component. The basic assumption of RUT is that every individual makes rational choices and maximizes his utility. Based on this assumption, it is possible to say that the alternative i will be preferred over another option j if and only if $Ui > Uj$. This condition is strongly associated with the probability of choice where each individual n will prefer option i over to other options j in choice set C which is written as follows:

$$Prob\left(i/_{C_n}\right) = Prob\left(V_{in} + \in_{in} > V_{jn} + \in_{jn}\right) for\ all\ j \in C\ and\ j \neq i \tag{2}$$

Eq. (2) holds the principle of RUT, which shows the stochastic components are identically and independently distributed (IID) with Gumbel or Weibull distribution. This led to use of the logit model, conditional logit and multinomial or ordered logit model with and without the alternative specific constant (ASC).

Based on the RUT and specified utility as a linear function of our proposed attributes for improved cyclone EWS and the respondents' SED characteristics, we developed respondents' indirect utility for improved cyclone EWS as follows:

$$U_i = ASC + \beta V_i + \gamma S_j + \varepsilon \tag{3}$$

where U_i indicates the indirect utility for program i, ASC stands for alternative specific constant or simple a constant, Z indicates a vector of program related attributes, S shows a vector of respondent's SED characteristics, and ε indicates the stochastic unobserved component.

In the study, four attributes such as accuracy of mean track error at the landfall time, updates information in advance, mobile phone-based warning, and increased additional annual payment to your household were set as categorical dummies and taken into consideration in six logit models. Furthermore, SED characteristics such as age, annual income, household size, and schooling years were continuous and multiplied with our proposed attributes in the extended model for measuring respondents' heterogeneous behavior toward preferences. Preference was a dichotomous dummy variable (status quo = 0, improved level = 1) in nature and treated as the outcome variable. All of our six regression models, three are estimated as conditional logit (CL) models and the rest three models are estimated as mixed logit (ML) models. Within each of CL and ML regression models, the first model was a standard model having no ASC, second model considered ASC for capturing the effect of unobserved determinants not included in the attribute vector, and the third model was the extended regression model with interaction terms to deal with respondents' heterogeneous preferences. Hypothetically improved levels were considered when ASC takes value 1and 0 for otherwise.

The payment attribute is connected with the other attributes to determine the MWTP, WTP and CS for measuring respondents' gains or losses and investor's revenue stream for improved EWS [54]. The value of MWTP is known as the implicit price or part-worth of the attribute. The general form of the MWTP in linear utility function is between the implicit price and other attributes. Total willingness-to-pay (TWTP) or simple WTP is calculated by summing the MWTP for each item. The mathematical form of MWTP can be written as follows:

$$MWTP_i = -\frac{du/_{di}}{du/_{dPayment\ attribute}} = -\frac{\beta_i}{\beta_{Payment\ attribute}} \tag{4}$$

To implement any kind of policy, we need to assess respondents' ability to pay for improved EWS. Their satisfaction is reflected by their WTP that helps to design and implement policy perfectly. The difference between the gains after implementation and before implementation of policy are treated as Consumer Surplus (CS) given by economist Alfred Marshall. The CS can be considered as utility received by the consumer in monetary terms by the choosing of one choice out of many. A rational individual or consumer always chooses the best option that provides the highest satisfaction or utility. The mathematical form of CS of individual n can be written mathematically as follows:

$$CS_n = \left(\frac{1}{\propto_n}\right) max_j\left(U_{nj}\forall j\right) \tag{5}$$

where $\propto_n$ indicates the marginal utility of income (presented by the coefficient of the payment) and it is equal to $\frac{dU_{nj}}{dY_n}$. The change in CS is defined as the difference between status quo and implemented program. It can be written as follows:

$$\Delta E(CS_{(n)}) = \left({}^{1}/_{\propto_n}\right)\left[ln\left(\sum_{j=1}^{J^1} e^{V_{nj}^1}\right) - ln\left(\sum_{j=1}^{J^0} e^{V_{nj}^0}\right)\right] \tag{6}$$

Perfect calculation of WTP and CS always require perfection of experimental design of choice. An experiment structured by the scientific terms includes the observation of the effect of response or outcome variable, given the manipulation of the levels of one or more other explanatory variables. The econometric software NLOGIT 4.0 was used to estimate the parameters and MWTP, WTP, and CS.

The CE approach consists of four basic steps. Each step depends on the previous [55]. The first step defines proposed attributes and their levels for constructing the choice card. The second step develops an experimental design by selecting the model and defining utility function. The third step, develops a survey questionnaire. This step has a few warm up activities such as a choice card for WTP, SED characteristics of respondents and proper sampling strategy, and the fourth step covers data input and analysis.

The selection of attributes and levels are key components in the choice experiment design [67]. To fulfill the research objectives, the attributes are used to describe a cyclone EWS to the respondents. The attributes will be the observed explanatory variables in the estimation process of our regression models. Customization is an issue in the selection of the pecuniary attribute and its level. Under this system, it should attempt to make choice alternatives more realistic by covering pecuniary attributes and its level [51]. Level can be expressed qualitatively, and the quantitative attributes can be presented in absolute or relative terms [56]. We considered the simplicity of the attributes in terms of meaning and measurement. If the respondent cannot understand the meaning of the attributes, then the estimated result may cause confusion. For conducting reliable choice experiments, we depend on in-depth review of relevant literature at the first stage for selecting a set of attributes. This set of attributes was then distributed with nine local, regional and national disaster managers and practitioner, researcher of meteorology, meteorological scientist, meteorological expert, high official of the Bangladesh Meteorological Department (BMD), *Upazila* (sub-district) Project Implementation Officer (PIO), and volunteer of Cyclone Preparedness Program (CPP) in order to get relevant attributes of EWS based on their experience. Due to funding and time limitations, we used telephone and email to collect their opinions about necessary attributes of cyclone EWS across coastal Bangladesh. They suggested eleven attributes such as precise time of cyclone landfall with possible impacts, accuracy of tropical cyclone forecast, number of updates per day, mobile phone-based short message warning, annual payment for cyclone EWS, time of landfall accuracy, maximum wind speed accuracy, projected location of landfall accuracy, expected storm surge accuracy, accuracy of mean track error at the landfall time, and updates information in advance were selected from Ahsan et al. [40]; Nguyen et al. [22] and Lazo, Waldman, Morrow, and Thacher [57]. Later, four Focus Group Discussions (FGDs) were carried out to narrow down the list for attributes with their levels, and determine base price for cyclone EWS. These FGDs were organized on 19–23 April 2018 at four different *Upazil* in four coastal districts (Satkhira, Khulna, Bagerhat and Cox's Bazar) of Bangladesh. Participants from heterogeneous occupations and direct victims to cyclones Sidr (November 2007), Aila (May 2009), Mahasen (May 2013), and Roanu (May 2016) took part in the discussion. The participants were selected in a non-random fashion. Four attributes such as *accuracy of mean track error at the time of landfall, advance and frequency of updates information, warning through mobile phone-based SMS, and additional annual payment for additional EWS* and their levels were selected on the basis of experts' opinion and participants of our FGDs. There is a rule of thumb that a bigger difference in each level of attribute in the choice experiment may reduce respondents' interest to hypothetical goods and services and narrow down the scope of choice experiment [58]. The pre-tests were conducted on 7–20 May 2018 in Khulna and Satkhira districts to validate and appropriateness of the selected attributes and understand the experimental setup where 30 respondents were participating. These 30 respondents were omitted from the main survey results. Results of our pre-tests assured that our finally selected four attributes and their levels were significant and relevant in the pre-tests. Table 1 outlines our selected attributes and their levels.

As the coastal people at risk pondered an improved EWS, accuracy of mean track error at the time of landfall seemed to be the most important and therefore, they were concerned about this attribute [59]. Respondents in our pre-tests also stated that they prefer the accuracy of mean track error at the time of landfall because of its significant contribution to detect the path of cyclones. The greater frequency of updates indicates up-to-date information on the present status of wind

Table 1
Attributes and their associated levels.

No.	Attribute	No.	Levels	Description
1	Accuracy of mean track error at the landfall time	1	98 km	Accuracy of mean track error in km (Status quo)
		2	80 km	Accuracy of mean track error in km lower than 1
		3	60 km	Accuracy of mean track error in km lower than 2
		4	40 km	Accuracy of mean track error in km lower than 3
2	Updates information in advance	1	10 h	Update information in hours (Status quo)
		2	12 h	Update information in hours higher than 1
		3	14 h	Update information in hours higher than 2
		4	16 h	Update information in hours higher than 3
3	Mobile Phone SMS warning	1	Available	Available phone SMS warning
		2	Not available	Not available phone SMS warning (Status quo)
4	Additional annual payment to your household	1	100 Taka (US$ 1.19)	Annual fee for early warning service in Tk.
		2	90 Taka (US $ 1.07)	Annual fee for warning service in Tk. & lower than level 1
		3	75 Taka (US $ 0.89)	Annual fee for warning service in Tk. & lower than level 2
		4	50 Taka (US $ 0.56)	Annual fee for warning service in Tk. & lower than level 3

Note. Tk. Stands for Taka. It is the currency of Bangladesh. Rate of Exchange 1 USD = 84 BDT.

speed and new position that are essential for making prediction to forecast of cyclones [22]. Respondents of our pre-tests give more importance on frequency of updates and information on the likely impacts of the imminent hazard. Villagers of remote villages and isolated islands and fishermen of river and off-shore have limited access to cyclone EWS through television, community radio, and other traditional means for early warning. Volunteers of CPP and participants of FGDs argued that SMS is the most convenient way to disseminate cyclone warning. In addition, People in remote villages and isolated islands have access to mobile phone network services. Coastal people who do not have mobile phones can get updated cyclone information from neighbors, relatives, community leaders who have mobile phones. For fishermen catching fish in deep sea can use high frequency radio telephones that can be used to contact relatives living in mobile network service areas in coastal areas [22]. All the improvement levels of our proposed attributes were chosen in terms of opinions of meteorological scientists and experts except the levels of payment attribute. Our collected Information on respondents' attitudinal parameters is not concerned to unveil determinants affecting WTP for improved EWS in Bangladeshi coast but to assess the investment prospect for our proposed improvements in the same region. A payment attribute plays an important role to achieve this goal. We proposed a compulsory annual payment mechanism. A payment attribute is considered to be that choice task for measuring MWTP, WTP and CS respectively [57]. Based on the findings of FGDs, we set four levels of payment.

3.2. Choice card design

Opinions of experts and findings of FGDs, and pre-tests were sequentially incorporated into the final design of our CE. In our study, the number of combinations of hypothetical attributes and their levels become too large $(4^3 \times 2^1) = (4X4X4X2) = 128$ by using *R* software. To reduce the cognitive burden on respondents, we apply orthogonal design principle using fractional design and reduce it randomly to 16 for 4 versions of the questionnaire where each version contains 4 combinations of choice at present conditions of 6 combinations for two choice cards. According to [67], respondents are not willing to consider more than five choice sets on average in the survey period. From the two choice cards, each respondent is requested to select two cards from two sets separately and randomly and answered twice. In the first selection of a card from the first set, each respondent gets 3 options such as option A, option B, and status quo. After selection of the first card, he selects another card from the second set consisting of another 3 options. Option A and option B show the potential improvement program and status quo presents the present condition.

The CE is affected by the anchoring and ordering effects, sampling error, and hypothetical bias due to familiarity with individual choice, order of improvement levels of choice tasks, wrong understanding of the objective of the survey and wrong imagination on hypothetical attributes. To avoid such situations, Firstly, we set four different price levels for each improvement to overcome the problem of anchoring effect. Secondly, data collectors randomly displayed six choice cards to each respondent and requested him to select two choice cards and ensured him the same choice cards were not previously presented to two consecutive respondents. Thirdly, a questionnaire of our study designed to explain the purpose of the survey. All respondents were briefed the importance of EWS by the data collectors. Lastly, all the data collectors interpreted the proposed attributes of EWS with practical demonstration. Data collectors allowed adequate time to respondents for proper responses from them through questionnaire and choice experiment. The data collectors did not indulge in personal and unnecessary gossiping to avoid anchoring and influencing the answers of the respondents. Each respondent was allowed to complete one survey only and twice selection of choice cards.

3.3. Questionnaire design, sampling technique and data collection

Extensive body of empirical works on EWS, opinions from experts, and pre-test helped to design our questionnaire. The questionnaire has three sections. The first section highlighted the SED characteristics of respondents. The second section presented the cyclonic hazards, frequency, and magnitude. The third or final section included CE eliciting respondents' WTP and CS for improved EWS.

It is difficult to determine the actual or optimal sample size for non-linear choice models because it depends on the true values of the unknown estimated parameters [55]. Train [60] argues that if the sample is exogenously drawn (either random or stratified random), maximum likelihood estimation may be used. If the sample is endogenously drawn, conditional maximum likelihood estimation is essential. In this study we used an exogenously stratified sampling method for sample selection and the villages were used as the strata. The target sample size was fixed at ($n =$) 225 followed to Lazo and Waldman [61]. Every respondent was requested to select two cards from two sets separately and randomly and put a tick marks twice in the preferred attributes of drawn two cards that makes the sample size double (225 X 2 =) 450. The final sample size included in the analysis is 450. This double counting practice of sample size is supported by Schaafsma, Brouwer, Liekens, and De Nocker [62]. *Upazila Nirbahi* Officers (UNOs) (Chief Executive of the smallest administrative unit of Bangladesh), Project Implementation Officers (PIOs) in *Upazila* (sub-district), volunteers of Cyclone Preparedness Program (CPP), and Chairman of different local Union Councils helped to design sampling setting.

Surveys through questionnaires were undertaken in a few coastal villages of Satkhira, Khulna, Bagerhat and Cox's Bazar districts between June and September 2018 (see the four circles in the study locations map for more details). These four coastal districts are affected at different magnitude levels of cyclones almost every year. Household in the study location was treated as a sample unit and a household member who was over 18 years old requested a response to the questionnaire. The authors

and other paid 4 data collectors were involved in the whole survey process to collect SP data. The respondents interviewed were either heads of household or spouses of the heads of household. Out of 352 respondents, 225 (63.92%) agreed to participate in the survey and CE and the remaining 127 (36.08%) respondents were refused the request to participate in the survey and CE.

Agreed respondents were asked whether they would like to pay for EWS. Of these, 83 respondents (36.88%) replied negatively due to low income and effective utilization of collected funds. Though all the respondents appreciated the proposed improvement intervention, 24 respondents among 83 of no replied respondents opined that improved EWS were not their concern and finance should come from the Government level. They were not interested in paying for improved EWS. Thus, they were considered as protest bidders. We therefore considered responses of 59 out of 83 respondents as valid zero.

Sample representativeness was explored by comparing SED characteristics with the last available Bangladesh Census data 2011. Table 2 shows the dominant SED characteristics in the study. Respondents' latitude and longitude related data were recorded using a Global Positioning System (GPS). All data collection protocols were approved for compliance with Human Subject research.

4. Result and discussion

4.1. Socioeconomic and demographic profile of the respondents

Our final sample included 209 respondents after screening for completeness. We found our sample was skewed toward greater representation of individuals with a lower education. Most of our samples have lower income, less education and to have a larger household size. In this study, the respondents are of an average age of 35.51 years and 22% are female. Of the respondents, 41% are illiterate, 27% completed primary school, 16% completed secondary school, 11% completed higher secondary school, and 5% complete their graduation. Years of schooling is recorded at 5.67 implies that the majority of respondents have completed their primary education. The average household size is 5.67 and the average annual income is BDT 103,508 (US$ 1232).

4.2. Econometric estimation

The econometric results of all the logit regression (three of them are CL and the rest three are ML) models are presented in Table 3. All of our proposed attributes are assumed to be random parameters and normally distributed because of heterogeneous preference which allows us to estimate standard deviation (SD). We measured the values of Log-likelihood (LL), Pseudo *R*-square, and Akaike Information Criterion (AIC) because of their overall appropriateness of the estimation whether the model is significant or not. Value of LL closer to zero represents a decent model fit for choice experiment analysis [41]. Similarly, the value of Pseudo *R*-square greater than 0.2 represents a well fitted model.

Three issues come to attention in the process of model selection such as coding of explanatory variables, inclusion of interaction of SED characteristics with proposed attributes and inclusion or exclusion of ASC. Appropriate coding strategy such as numerical attributes, effects coding and dummy coding is essential for measuring the parameters, avoiding correlation with the ASC and minimizing collinearity. The ASC along with proposed attributes and variables represent the variation of outcome variable essential for measuring welfare. Interaction of SED characteristics with proposed attributes is essential for controlling heterogeneity in preferences toward any improved programs. Thus, we considered all these three issues in our model.

Table 2
Summary statistics of major SED characteristics of sample households.

Variables	Maximum	Minimum	Mean	Std. dev
Age	73	18	34.51	14.12
Household size	11	4	5.17	1.87
Schooling years	16	0	5.67	3.96
Annual income	153910	20000	103, 507.52	22604.17

The estimated coefficients of all attributes of CL in the standard model are positive and statistically significant at conventional levels except payment. The positive sign of all proposed attributes implies that coastal households have higher demand for improved EWS. On the contrary, the negative sign of payment is consistent with the traditional demand theory i.e., increase in payment of EWS, keeping other attributes constant, would shrink demand for EWS and vice-versa. The negative payment value indicates that households would have to be compensated per year for the improved EWS, if their utility is to be held constant [54]. Similarly, estimated coefficients of all attributes of the second CL model with ASC and third CL model with ASC and interactions terms are also positive and significant. These estimated coefficients also support the coastal households' preferences for improved EWS. Three out of nine interacted SED characteristics with proposed attributes in the third CL extended model expose significant relation with the respondents' preferences toward improved EWS. The third CL model seems to be more appropriate compared to other CL models because of its controlling heterogeneity in preferences toward EWS.

The estimated coefficients of all attributes of ML models yield similar results. Estimated results of log-likelihood, Pseudo *R*-squared and AIC make a guarantee that all of the ML models are more efficient compared to those of all CL models. Furthermore, two out of nine interaction terms are found to be statistically significant. Whereas, five out of nine interaction terms are found to be statistically significant. These results also support the efficiency of all ML models compared to those of all CL models. Positive values of AIC in the second and third MNL models and inputting interaction terms in the third ML model display more robust results.

It is noticed that for every increment from the lower level to immediate upper level, all proposed attributes are positively and significantly affected respondents' preferences for improved EWS. Additionally, the age of the respondents have a more likely negative effect on the delivery mode of improved EWS across coastal Bangladesh. For instance, negative sign of estimated coefficients of interaction of accuracy of mean track error in the landfill time-age and SMS-age exhibit more likely negative effect on the preferences for improved EWS. These findings suggest that adult members in the coastal households are less likely to prefer accuracy of mean track error in the landfill time and SMS instead they are more preferred to receive frequent voice messages over mobile phone. This finding is consistent with the study by Ahsan et al. [40]. The underlying facts of such findings imply that a major portion of adults in coastal households are illiterate. They cannot read SMS on their mobile phones properly. In addition, all adults have less tendency to read every SMS because of the crowding out effect that occurs in a large number of commercial SMS every day. Meanwhile, older adults are familiar and more experienced with cyclonic events and other unexpected hazardous conditions and they can predict the likelihood of intensity of cyclones by observing natural signals and signs [40].

4.3. MWTP, average WTP and WTP estimation

The estimated results of CL and ML models along with the upper and lower bound results are presented in Table 4.

Positive values of MWTP at all levels indicate that coastal households have positive preferences toward improved EWS. The standard model in CL shows sampled respondents have highest MWTP for SMS, which is estimated at BDT 27.146 (US$ 0.323) more with each level of improvement in terms of the status quo. The model with ASC yields almost the identical MWTP, while the model with interaction shows a slightly decrease in MWTP for each level of improvement. On the contrary, the model with interaction in ML also exhibits the higher MWTP for the same attribute, which is calculated at BDT 26.548 (US$ 0.316).

Table 3
Results of estimated regression models for improved EWS.

Conditional logit (CL)				Mixed logit (ML)					
Variables	Standard model	Model with ASC	Model with interaction	Standard Model		Model with ASC		Model with interaction	SD
	Mean	Mean	Mean	Mean	SD	Mean	SD	Mean	SD
ASC		2.780*** (0.485)	2.572* (0.487)			5.053 (0.394)	2.521 (0.007)	3.440* (2.001)	3.453*** (0.651)
Track	0.009** (0.005)	0.010** (0.005)	0.017** (0.121)	0.006** (0.003)	0.446 (0.009)	0.008** (0.247)	0.467* (0.006)	0.011** (0.005)	0.253* (0.063)
Update	0.093* (0.055)	0.096* (0.055)	0.193* (0.122)	0.082 (0.032)	0.193* (0.003)	0.087** (0.001)	0.835 (0.012)	0.105* (0.055)	0.179 (0.004)
SMS	1.113*** (0.198)	1.112*** (0.198)	0.873** (0.670)	0.185* (0.134)	0.475* (0.194)	0.187*** (0.048)	0.575 (0.004)	1.115*** (0.198)	0.944** (0.364)
Payment	−0.041*** (0.014)	−0.041*** (0.004)	−0.039*** (0.013)	−0.079** (0.005)		−0.083** (0.007)		−0.042*** (0.004)	
Track*Inc			0.173 (0.185)					0.360 (0.289)	0.229** (0.005)
Update*Inc			0.053 (0.075)					0.075 (0.038)	0.023 (0.017)
SMS*Inc			0.138** (0.205)					0.654** (0.274)	0.071 (0.070)*
Track*Age			−0.229 (0.219)					−0.552 (0.216)	0.229
Update*Age			0.017 (0.030)					0.209** (0.185)	(0.001)
SMS*Age			−0.139**					−0.019**	0.362*
Track*Syrs			(0.007)					(0.006)	(0.005)
Update*Syrs			0.011 (0.153)					0.050 (0.217)	0.003
SMS*Syrs			0.224 (0.218)					0.042** (0.286)	(0.018)
			0.197 (0.103)					0.117*** (0.152)	0.623 (0.153)*
									0.227 (0.376)
									0.083 (0.004)
Log-likelihood	−347.9045	−341.6288	−822.0943	−404.8041		−403.0093		−335.0783	
R-square	0.217985	0.256067	0.220949	0.318765		0.314032		0.37033	
AIC	2.893091	1.66808	2.964301	1.894302		1.783675		1.66545	
Prob > Chi^2	0.000000	0.000000	0.000000	0.000000		0.000000		0.000000	

Note. Inc indicates annual income, Syrs indicates schooling years.
Observation n = 438 (for 219 respondents) because of double tick marks in the preferred attributes of drawn two cards.
***$p < 0.10$; **$p < 0.05$; *$p < 0.001$; Robust standard errors are placed in parentheses.

Table 4
Estimation of MWTP (in BDT).

Attributes	CL model			ML model		
	Standard model	Model with ASC	Model with interaction	Standard model	Model with ASC	Model with interaction
Track	0.219	0.243	0.435	0.075	0.096	0.261
Lower bound estimate	0.207	0.139	0.013	0.001	0.003	0.117
Upper bound estimate	0.228	0.299	0.758	0.178	0.885	1.032
Update	2.268	2.341	4.949	1.038	1.048	2.500
Lower bound estimate	1.198	1.109	1.433	0.350	0.574	1.098
Upper bound estimate	2.302	4.439	7.298	2.056	3.044	3.728
SMS	27.146	27.121	22.385	2.341	2.253	26.548
Lower bound estimate	21.432	13.294	17.004	1.998	1.953	17.375
Upper bound estimate	29.013	32.606	30.373	2.670	4.226	28.942

The standard model and model with ASC in ML calculate lower and almost identical MWTP. The subsequent higher MWTP is observed in a model with interaction in CL for update information, which is recorded at BDT 4.949 (US$ 0.059). The standard model and model with ASC calculate almost the same MWTP. On the other hand, models with interaction in ML estimates higher MWTP for the same attribute compared to those of standard model and model with ASC in ML. Among all the improvement attributes, the least MWTP is observed in accuracy of mean track error at the landfall time. The model with interaction in CL exhibits the estimated value of MWTP for the same attribute recorded at BDT 0.435 (US$ 0.005). While MWTP in the standard model and model with ASC in CL yields the lower and same results. The lowest value of MWTP is recorded at BDT 0.075 (USD 0.0009) in the standard model in ML. Table 5 outlines the values of average willingness-to-pay (AWTP) of our proposed attributes and WTP. Calculation of AWTP is derived from the model with interaction in ML because of its excellency and efficiency compared to those of other regression models in Table 3.

Estimated result of WTP from mean level attributes suggests that

Table 5
Estimation of AWTP and WTP (in BDT).

Attributes	AWTP at mean attribute level
Track	10.99
Update	19.48
SMS	31.90
WTP	62.37

coastal households are interested to pay BDT 62.37 (US$ 0.743) in a year for obtaining improved EWS. Estimated annual average income BDT 103,508 (US$ 1232) of sampled respondents implies that WTP for improved EWS is about 0.06% of their yearly income. This empirical assessment is consistent with the study by Nguyen et al. [22] and Lazo et al. [57] as they reported that WTP of sampled households in Vietnam is about 0.024–0.045% and it is around 0.024–0.045% in the United States. Meanwhile, Ahsan et al. [33] estimated WTP is 0.46% of yearly average income across coastal Bangladesh. The higher percentage of WTP may be the result of higher level of payment in their choice card.

4.4. Estimation of annual CS and revenue stream

The WTP is the maximum amount that a household will pay for a desirable set of services like improved EWS. At a payment equal to WTP, the household would be indifferent about paying the services. On the other hand, at a payment smaller to WTP, the household would be gained. This is the concern of CS. It shows the discount rate free amount of money where a household is ready to pay for a set of desirable services minus the amount the household actually pays for it. Based on the ML model in Table 3 and using Eq. (6), the estimated value for different CS is presented in Table 6.

The measurement of economic benefits from consumer's viewpoint involves an investigation of the difference between the preference of the consumer that could be achieved under the status quo and changed scenario alternatives [64]. We proposed three alternative improvement scenarios for improved EWS in coastal Bangladesh. The minimum improvement option one offers is that the accuracy of cyclone prediction will be increased by reducing track error about 80 km, and update information will be provided 12 h in advance by mobile phone-based service. The moderate improvement option two offers the accuracy of cyclone prediction will be increased by reducing track error about 60 km, and update information will be provided 14 h in advance by mobile phone-based service. The maximum improvement option 3 offers the accuracy of cyclone prediction will be increased by reducing track error about 40 km, and update information will be provided 16 h in advance by mobile phone-based service. Estimated results ensure that each household annually gets more surplus BDT 127.06 (US$ 1.512) from option three.

Based on the model with interaction in ML in Table 3, monetization of the investment prospects for EWS across coastal Bangladesh is undertaken. Statements of BBS [65] and Dasgupta et al. [66] motivated us to assess the investment prospects for improved EWS. BBS [65] reports that total coastal households are recorded at 8,712,619. Dasgupta et al. [66] point out that the number of vulnerable people in coastal Bangladesh would be 6.03 million by the year 2030 because of negative impacts of extreme climatic events. Table 7 outlines the yearly revenue stream derived from the payment for improved cyclone EWS.

Investment in improved cyclone EWS in coastal Bangladesh makes a guarantee to earn revenue BDT 543.406 million (US$ 6.469 million) in a year. If an investor invests 10 years of a project for improved EWS, the future stream of revenue at 5% discounting rate would be BDT 5.46 billion (US$ 0.065 billion). An investment in improved cyclone EWS appears to be effective, feasible and economically sustainable. Based on this estimation, an investor not only gets returns from such project but also protects property, livelihood and lives from unexpected extreme cyclonic events.

5. Conclusions

Climate change-induced disaster has no longer scientific speculation about its certainty. Adaptation measures along with disaster preparedness is highly required for minimizing the damage generated from cyclones. Improved EWS could be a potential contributor to adaptation that minimizes loss and damage from cyclonic events across coastal Bangladesh. Considerable capital investment can accelerate improved EWS. Our study attempts to provide guidelines for designing means of economic valuation for improved cyclone EWS through MWTP, AWTP, WTP, CS and PV. We applied SP data and CE for valuing and setting desirable attributes and their levels.

Interaction of coastal households' SED characteristics with proposed attributes are essential for getting significant results and reducing heterogeneity problems. Coastal households are willing to pay for the improved EWS over the existing EWS. Sampled respondents have the highest MWTP for SMS, which is estimated at BDT 27.146 (US$ 0.323) more with each level of improvement in terms of the status quo. WTP from mean level attributes suggests that coastal households are interested to pay BDT 62.37 (US$ 0.743) in a year for obtaining improved EWS. Average income BDT 103,508 (US$ 1232) of sampled respondents implies that WTP for improved EWS is about 0.06% of their yearly income. This assessment is consistent with the study by Nguyen et al. [22] in Vietnam and Lazo et al. [57] in the United States. The maximum improvement option three offers the accuracy of cyclone prediction will be increased by reducing track error about 40 km, and update information will be provided 16 h in advance by mobile phone. Each coastal household gets more surplus BDT 127.06 (US$ 1.512) from the maximum improvement option. Value for revenue stream BDT 5.46 billion (US$ 0.065 billion) confirms that investment in improved cyclone EWS in coastal Bangladesh ensures more return in a year. This assessment is consistent with the study by Ahsan et al. [40].

We applied CE for valuing our proposed attributes for cyclone EWS by different assessment tools and techniques like MWTP, WTP, CS, and PV. As our study did not consider voice message in local dialects, maximum wind speed accuracy, and other common natural hazards, there is a greater scope to consider such attributes and common natural hazards with larger sample size and greater coastal regions for further study. In addition, the contingent valuation method can also be an effective method for estimating MWTP and WTP. Our future work will expand the current work by considering voice message in local dialects, maximum wind speed accuracy, floods and droughts, larger sample size and contingent valuation method.

Table 6
Estimation of CS for different options for EWS.

Policy scenario	CS (in BDT)
Option one	106.10
Option two	116.57
Option three	127.06

Table 7
Yearly revenue stream by PV for improved EWS.

WTP (in BDT)	Number of coastal household	Annual revenue (in BDT)	Annual revenue (in US$)
62.37	8,712,619	543,406,047	6,469,120

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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